

## MOTIONCORE · DESIGN DECISION RECORD

# MotionCore recommendations accepted

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## DOCUMENT CONTROL

|             |                                                               |         |                |
|-------------|---------------------------------------------------------------|---------|----------------|
| Document ID | MTC-DDR-002                                                   | Version | 0.1            |
| Status      | <b>Draft</b>                                                  | Date    | 2026-09-25     |
| Author      | Amish Chadha                                                  | License | CERN-OHL-S-2.0 |
| Project TRL | <b>TRL 3</b> Proof of concept                                 |         |                |
| Repository  | github.com/BoujeeEnjinia1701/motioncore                       |         |                |
| Source      | docs/decisions/0002-recommendations-accepted.md @ uncommitted |         |                |

## REVISION HISTORY

| Version | Date       | Author       | Change                                                                                                                          |
|---------|------------|--------------|---------------------------------------------------------------------------------------------------------------------------------|
| 0.1     | 2026-09-25 | Amish Chadha | Recommendations accepted by Amish (DDR-002); record every newly decided item, what changed in the repo and the items still open |

## 0002: Recommendations accepted

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- **Date:** 2026-09-25
- **Status:** accepted

### ■ Context

On 2026-09-25 Amish wrote: "i accept all your recommendations, go with them across all repos." Before that, MTC-DDR-001 had adopted the eleven TRL 2 recommendations for TRL 3 work, open for his review, and the TRL 3 review note ( docs/REVIEW.md ) listed further items as proposed, awaiting Amish. This record lists every item that the instruction turns into a decision, what changed in this repo because of it, and the items that remain open because no recommendation was made. The portfolio stays at TRL 3; TRL 4 is on hold by Amish's instruction.

### ■ Options considered

The options for each item are in docs/REVIEW.md (both sessions of 2026-09-25), MTC-DDR-001 and MTC-CAL-001. Where a recommendation named one of several options, that option is the decision.

### ■ Decision

*Table 1. Items decided by Amish, 2026-09-25: go with recommendation.*

| #       | Item (source)                                                                                                                                                                                                                                    | Decision                                                                                                                                                                                                                                        | What changed in the repo                                                                                                                                                                                                                                                                                                                                                   |
|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 to 11 | TRL 2 review items: budget scope, 20 V lower bound, VESC-class controller, DC contactor, stop category 0, interface v0.1, SwapCell heartbeat, CellGuard link, 350 W heavy case, separate firmware, CargoMule as first host (MTC-DDR-001 Table 1) | Decided as recommended; they were already built into the TRL 3 design                                                                                                                                                                           | Wording only: "adopted for TRL 3, open for review" replaced by "decided by Amish" in MTC-DDR-001 v0.2, MTC-PRB-001 v0.4, MTC-PRC-001 v0.4, MTC-REQ-001 v0.4 and <code>bom/bom-notes.md</code> . <code>budget_usd</code> stays \$300 (item 1 redefined what the budget covers; no new figure was recommended)                                                               |
| 15      | R12 mass shortfall (MTC-DDR-001 item 15)                                                                                                                                                                                                         | Option (a): relax R12 from 1.5 kg to 2.0 kg, because the enclosure is the heat sink and R9 is already at risk                                                                                                                                   | MTC-REQ-001 v0.4 R12 target 2.0 kg; R12 status from <b>not met</b> to <b>met</b> (1.94 kg, 0.06 kg margin); MTC-CAL-001 v0.2 and <code>sizing.py</code> ; drawing note on MTC-DWG-001 Rev P2; key figure on concept sheet MTC-DWG-010 Rev P3; README                                                                                                                       |
| 16      | TRL 3 engineering proposals from MTC-CAL-001 (MTC-DDR-001 item 16)                                                                                                                                                                               | Confirmed: two CAN buses on the supervisor, precharge closure at 0.6 s with a 90 % bus check, DC-rated fuses with 1 kA breaking capacity, 4 mm <sup>2</sup> pack leads for 24 V hosts, 24 V Zener coil suppressor, controller overvoltage fault | Already in <code>bom/bom.csv</code> and MTC-PRC-001; now listed as decided design rules (MTC-PRC-001 v0.4 item 11). Because of item 17, the fuse rating moves from 58 V DC to 60 V DC or more and the overvoltage fault from 60 V to 66 V                                                                                                                                  |
| 17      | R1 upper bound for CellGuard 16S LiFePO <sub>4</sub> packs, 58.4 V full (cross-repo note, MTC-DDR-001 item 17)                                                                                                                                   | Raise R1's upper bound from 58 V to 60 V                                                                                                                                                                                                        | MTC-REQ-001 v0.4 R1 20 to 60 V; MTC-CAL-001 v0.2: overvoltage fault 66 V, bus peak after a regenerating contactor opening 62.0 V to 67.8 V (inside 75 V), direct-drive back-EMF limit 1.29 to 1.25 times no-load speed, precharge peak 0.60 A at 60 V; <code>bom/bom.csv</code> items 2 and 5; MTC-PRB-001 v0.4; MTC-PRC-001 v0.4; drawing and concept sheet notes; README |
| 18      | R10 sealing (MTC-DDR-001 item 18)                                                                                                                                                                                                                | Evaluate a sealed power connector before interface v0.1 is frozen                                                                                                                                                                               | The XT90 is marked provisional in MTC-PRC-001 v0.4 Table 2, <code>bom/bom.csv</code> item 8, MTC-DWG-001 Rev P2 and MTC-CAL-001 v0.2. The geometry is unchanged until a connector is chosen. R10 stays at risk. The evaluation itself is a paper task for a later TRL 3 session                                                                                            |

**Consequence of combining items 16 and 17.** Item 16 confirmed a 60 V controller overvoltage fault, and item 17 raised the supply range to 60 V. Both cannot hold: a full CellGuard 16S pack (58.4 V) would sit 1.6 V below the fault. The overvoltage fault therefore moves to 66 V, which keeps the bus under 75 V after a contactor opening during regeneration (67.8 V), and the fuse rating follows R1 to 60 V DC. These two values are engineering consequences of Amish's decisions, recorded here so he can see them.

No design change alters geometry, mass or price: `cad/src/model.py` is unchanged and the STEP and STL files were re-exported only to confirm the model runs. The MotionCore kit stays at \$265 against \$300, and `trl` and `trl_target` stay at 3.

## Items still open

Table 2. Items that remain proposed, awaiting Amish.

| #  | Item                                                                              | Why it stays open                                      |
|----|-----------------------------------------------------------------------------------|--------------------------------------------------------|
| 12 | Dual-motor hosts (PalletPilot): two controllers on one supervisor, or two modules | No recommendation was made                             |
| 13 | Brushed motors (StepClimber): VESC DC mode or a different power stage             | No recommendation was made                             |
| 19 | Contact coil economizer (heavy case about 55.7 °C instead of 58.5 °C)             | A suggestion only in MTC-CAL-001, not a recommendation |

Item 14 (whether SunSpoke keeps its own SwapCell charge-host adapter) belongs to the SunSpoke project and is listed under cross-repo actions in `docs/REVIEW.md`.

## Decided but on hold (TRL 4)

Testing of the safety functions, a bench build, the timed fit for R11 and any vibration test remain TRL 4 work and are on hold by Amish's instruction. No item above required TRL 4 work to be recorded.

## ■ Consequences

- Requirement status on paper: ten of fourteen met, none not met, three at risk (R2, R9, R10), one not verifiable at TRL 3 (R11).
- The prospective short-circuit current of a CellGuard 16S pack is not known in this repo; it must be checked against the 1 kA fuse breaking capacity before a CellGuard host is fitted.
- CellGuard, SunSpoke and GravitySort need follow-up in their own repos (cross-repo actions in `docs/REVIEW.md`). No other repo was edited.
- Record any later change to these items as a new decision record.